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Natural convection heat transfer modeling by the cascaded thermal lattice Boltzmann method

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ABSTRACT

In this article, we present application of the Cascaded Thermal Lattice Boltzmann Method (CTLBM) in simulations of natural convection in differentially heated square cavity with adiabatic top and bottom walls. This classical benchmark problem is solved for wide range of Rayleigh numbers (10^6 – 10^{10}) and compared with data from the literature. For high Rayleigh numbers we present comparison of Nusselt number and wall shear stress distributions along hot wall with experimental and direct numerical simulation (DNS) data. Results for Rayleigh numbers up to 10^6 are also compared with previous results obtained by MRT-LBM simulations of Wang et al. The results are in good agreement with the existing ones obtained numerically and experimentally.

1. Introduction

Lattice Boltzmann methods have established themselves as the viable alternative among numerical methods used in CFD. LBM solves the discretized Boltzmann Transport Equation (BTE) to obtain set of distribution functions (DFs), from which macroscopic quantities (density, pressure, velocity, temperature) are then obtained. The physics solved by the LBM is controlled by the form of the collision operator and chosen equilibria for the DFs. Several types of realization of collision operators have emerged [10,22,24,53]. The simplest collision operator is the Single Relaxation Time (SRT) sometimes called BGK in the LBM community, after the authors Bhatnagar, Gross and Krook [44]. A large number of heat transfer and fluid flow problems solved by SRT LBM are reported in the literature [5,7,11]. In the SRT approach, all non-conserved moments relax to their equilibrium with the same relaxation time (due to the construction of the collision operator). SRT is based on the BGK approach [22], which could produce numerical instabilities, when the lattice resolution is insufficient [40] and also the truncation error control is limited [56,57]. In order to increase the stability and accuracy of the LBM schemes, Multiple Relaxation Times (MRT) methods were proposed [10]. In MRT schemes collisions are performed in moment space and different moments could be relaxed with different relaxation times. MRT methods performed reasonably well and showed greater stability and accuracy compared to SRT LBM. Unfortunately the MRT methods are unstable for high Re flows and have other problems mentioned in Ref. [25]. The CLBM are methods where central moments are relaxed in a “cascaded manner“ [23,30]. CLBM was successfully used for high Re fluid flow and general heat transfer problems [2,18,19,21,24,37,46].

Some authors use MRT methods for the fluid flow and e.g. finite differences to solve energy equation independently [6,45]. This approach is known as Hybrid LBM. Other approach known as Double Distribution Function (DDF) scheme was proposed by Ref. [42], here two sets of DFs are used, one for the Navier-Stokes equations and another for the energy equation. A large number of research articles appeared, which describe DDF LBM approach with SRT and MRT LBM (see e.g. Refs. [40,52] and references therein). Recently, D_3Q_{27} DDF cascaded LBM was used for steady velocity field and solute transport in porous medium [20]. Another article describe CLBM scheme for the fluid flow and SRT LBM for the energy equation [21]. The double cascaded DDF LBM scheme for thermal problems was recently presented by the authors in Ref. [19], where CLBM-CTLBM approach was derived and applied to solve forced convection, meanwhile Fei et al. published somehow similar DDF CLBM approach applied to heat transfer [46].

In the present article we solve natural convection in a differentially heated cavity by the cascaded DDF LBM. We compare results obtained from our CTLBM code with data from literature. Flow and heat transfer

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in the square cavity for wide range of Rayleigh numbers have been studied by various groups and substantial research has been carried out. Various authors used finite differences, finite volumes, finite elements and pseudo-spectral methods [8,9,15,28,32–34], and also LBM [3,13,14,16,17,36,43]. Some of the researchers have adopted Hybrid LBM and DDF LBM [14]. The SRT [3,16] and MRT LBM were used for laminar flow regime [13,17,36,43]. Dixit et al. [16] used DDF LBM for solving internal energy equation along with counter-slip boundary conditions together with mesh refinement and simulated high Rayleigh number flows (up to $Ra = 10^{10}$) with SRT. Recently, Jami et al. [27] published a paper where two MRT DDF LBM schemes have been used to solve natural convection up to $Ra = 10^8$. Allen and Reis derived moment based boundary conditions and incorporated them in the MRT LBM in order to solve natural convection in a square cavity [36]. Ren et al. [35] presented the CUDA implementation of DDF LBM scheme with a SRT collision operator to solve natural convection in a square cavity with solid obstacles. Wang et al. [17] used MRT LBM to qualitatively examine natural convection in square cavity up to $Ra = 10^6$ and also Rayleigh-Bénard convection. High Ra number flows in cavities with aspect ratio close or equal to 1 were studied mainly by LES and DNS e.g. Refs. [49–51].

2. Double distribution cascaded lattice Boltzmann method for natural convection

DDF LBM approach solves two Lattice Boltzmann Equations, one describes evolution of velocity distribution functions and other describes evolution of temperature or internal energy distribution functions. They are solved simultaneously on two separate lattices. In this article, we consider approach where the temperature is considered as a passive scalar because viscous dissipation of heat and compression work are negligible. We use D_2Q_9 lattice model for fluid flow and D_2Q_5 lattice model for temperature field. The characteristic velocities of the lattice models are depicted in Fig. 1 and defined by following sets: $\mathbf{c}_i = (c_{i,x},\ c_{i,y})$ for D_2Q_9 lattice ($i = 1,\ ...,9$) are

$$\{(0,0),\ (-1,1),\ (-1,0),\ (-1,\ -1),\ (0,\ -1),\ (1,\ -1),\ (1,0),\ (1,1),\ (0,1)\},$$

and for D_2Q_5 lattice ($i = 1,\ ...,5$) are

$$\{(0,0),\ (-1,0),\ (0,\ -1),\ (1,0),\ (0,1)\}.$$

The weight factors, w_i for D_2Q_9 and D_2Q_5 are $w_1 = 4/9,\ w_{3,5,7,9} = 1/9,\ w_{2,4,6,8} = 1/36$ and $w_1 = 1/3,\ w_{2,3,4,5} = 1/6$, respectively.

2.1. Cascaded LBM for the flow field

The fluid behavior at mesoscopic scale is described by fluid particles in the framework of the Boltzmann's work and their properties at certain space and time are defined by moments $m_{\alpha\beta}$ of velocity distribution functions (DFs) $f(\mathbf{x}, \boldsymbol{\xi}, t)$. The evolution of such DFs obeys Boltzmann Transport Equation which reads

$$\frac{\partial f(\mathbf{x}, \boldsymbol{\xi}, t)}{\partial t} + \boldsymbol{\xi}\cdot\nabla f(\mathbf{x}, \boldsymbol{\xi}, t) = \Omega(f, f), \tag{1}$$

where ξ is the microscopic velocity, Ω is the collision operator. The spatial and temporal derivatives in BTE (1) are discretized, the velocity distribution functions are reduced to finite given by the desired lattice model, in our case the D_2Q_9 and D_2Q_5. Then by choosing the cascaded form of collision operator and incorporating forcing term $\widetilde{F_i}$ we end up with the cascaded lattice Boltzmann equation (CLBE), which in lattice units reads

$$f_i(\mathbf{x} + \mathbf{c}_i,\ t + 1) = f(\mathbf{x},\ t) + \mathbb{K}\cdot\mathbf{k} + \widetilde{F_i}, \tag{2}$$

where f_i is the "velocity" distribution function linked to the ith characteristic velocity, $\mathbb{K}$ is transformation matrix, $\mathbf{k}$ is a vector of moments of f_i resulting from the cascaded collision operator. The equilibrium distribution function f_i^{eq} is defined based on the Maxwell-Boltzmann equilibrium distribution function by

$$f_i^{eq} = \rho w_i\left(1 + \frac{\mathbf{u}\cdot\mathbf{c}_i}{c_s^2} + \frac{(\mathbf{u}\cdot\mathbf{c}_i)^2}{2c_s^4} - \frac{\mathbf{u}\cdot\mathbf{u}}{2c_s^2}\right), \tag{3}$$

where ρ is density, c_s is the speed of sound and $\mathbf{u} = (u,\ v)$ is macroscopic velocity vector. For the lattice model D_2Q_9 we have $c_s = 1/\sqrt{3}$. The body force can be modeled by different approaches, those are compared in Ref. [1] for the problem of a natural convection. We use the forcing scheme proposed by Ref. [58]. First the velocity field is modified by known force $\boldsymbol{\Gamma} = (\Gamma_x,\ \Gamma_y)$ (which is defined later),

$$u = \frac{1}{2\rho}\Gamma_x \quad v = \frac{1}{2\rho}\Gamma_y \tag{4}$$

then expressions for the components to be included in cascaded collisions are

$$\begin{aligned}
F_4 &= -(\Gamma_x u + \Gamma_y v)\\
F_5 &= -(\Gamma_x u - \Gamma_y v)\\
F_6 &= \tfrac{1}{2}(\Gamma_x v + \Gamma_y u)\\
F_7 &= -\tfrac{1}{2}(u^2\Gamma_y - \Gamma_x uv)\\
F_8 &= -\tfrac{1}{2}(v^2\Gamma_x - \Gamma_y uv)\\
F_9 &= -(\Gamma_x uv^2 + \Gamma_y u^2 v)
\end{aligned} \tag{5}$$

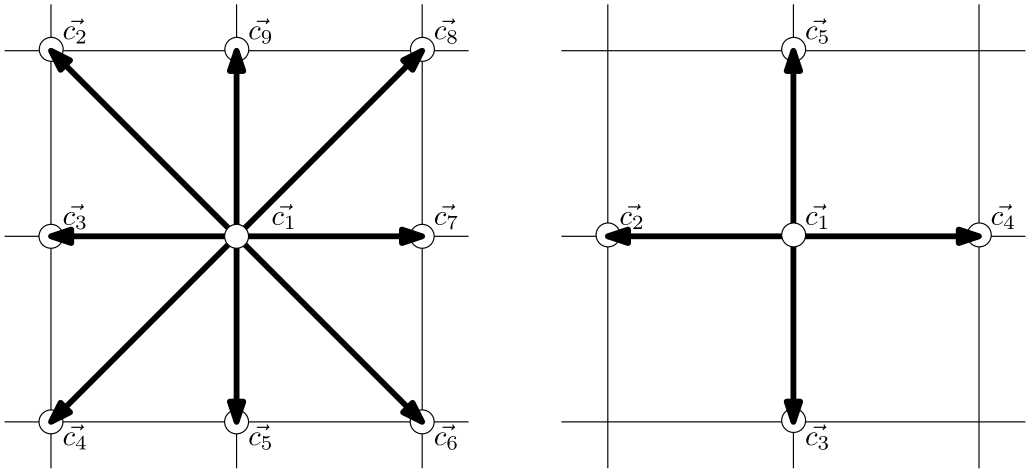


Fig. 1. Characteristic velocities (links) in 2D for D_2Q_9 (left) and D_2Q_5 (right) lattice models.

and the components which are added directly to the distribution functions are

$$\begin{aligned}
\widetilde{F_1} &= \tfrac{1}{9}(-\gamma_3+\gamma_8)\\
\widetilde{F_2} &= \tfrac{1}{36}(-6\gamma_1+6\gamma_2+2\gamma_3-9\gamma_5-3\gamma_6+3\gamma_7+\gamma_8)\\
\widetilde{F_3} &= \tfrac{1}{36}(-6\gamma_1-\gamma_3+9\gamma_4-6\gamma_7-2\gamma_8)\\
\widetilde{F_4} &= \tfrac{1}{36}(-6\gamma_1-6\gamma_2+2\gamma_3+9\gamma_5+3\gamma_6+3\gamma_7+\gamma_8)\\
\widetilde{F_5} &= \tfrac{1}{36}(-6\gamma_2-\gamma_3-9\gamma_4-6\gamma_6-2\gamma_8)\\
\widetilde{F_6} &= \tfrac{1}{36}(6\gamma_1-6\gamma_2+2\gamma_3-9\gamma_5+3\gamma_6-3\gamma_7+\gamma_8)\\
\widetilde{F_7} &= \tfrac{1}{36}(6\gamma_1-\gamma_3+9\gamma_4+6\gamma_7-2\gamma_8)\\
\widetilde{F_8} &= \tfrac{1}{36}(6\gamma_1+6\gamma_2+2\gamma_3+9\gamma_5-3\gamma_6-3\gamma_7+\gamma_8)\\
\widetilde{F_9} &= \tfrac{1}{36}(6\gamma_2-\gamma_3-9\gamma_4+6\gamma_6-2\gamma_8)
\end{aligned} \tag{6}$$

where γ_i are defined as

$$\begin{aligned}
\gamma_1 &= \Gamma_x\\
\gamma_2 &= \Gamma_y\\
\gamma_3 &= 6(\Gamma_x u+\Gamma_y v)\\
\gamma_4 &= 2(\Gamma_x u-\Gamma_y v)\\
\gamma_5 &= \Gamma_x v+\Gamma_y u\\
\gamma_6 &= (2-3u^2)\Gamma_y-6\Gamma_x uv\\
\gamma_7 &= (2-3v^2)\Gamma_x-6\Gamma_y uv\\
\gamma_8 &= 6((3v^2-2)\Gamma_x u+(3u^2-2)\Gamma_y v)
\end{aligned} \tag{7}$$

The macroscopic density ρ and macroscopic velocity **u** can be obtained as first two moments of f_i

$$\begin{aligned}
\rho &= \textstyle\sum_i f_i,\\
\rho\mathbf{u} &= \textstyle\sum_i \mathbf{c}_i f_i.
\end{aligned} \tag{8}$$

We define raw moments $m_{\alpha\beta}$ and central moments $\kappa_{\alpha\beta}$ of order $\alpha+\beta$ as

$$m_{\alpha\beta} = \sum_i c_{i,x}^{\alpha} c_{i,y}^{\beta} f_i, \tag{9}$$

$$\kappa_{\alpha\beta} = \sum_i (c_{i,x}-u)^{\alpha}(c_{i,y}-v)^{\beta} f_i. \tag{10}$$

From the above one can readily see that

$$\begin{aligned}
\rho &= m_{00} = \kappa_{00},\\
\rho u &= m_{10},\\
\rho v &= m_{01},\\
\kappa_{10} &= \kappa_{01} = 0.
\end{aligned} \tag{11}$$

To perform cascaded collision we need the transformation matrix $\mathbb{K}$ and the collision vector **k**. We choose following non-orthogonal moments for the flow field on the D_2Q_9 lattice (we use Einstein summation here)

$$\left\{\sum_i f_i,\ c_{i,x}f_i,\ c_{i,y}f_i,\ (c_{i,x}^2+c_{i,y}^2)f_i,\ (c_{i,x}^2-c_{i,y}^2)f_i,\ c_{i,x}c_{i,y}f_i,\ c_{i,x}^2c_{i,y}f_i,\right.$$
$$\left. c_{i,y}^2c_{i,x}f_i,\ c_{i,x}^2c_{i,y}^2f_i\right\}.$$

The first three moments are the conserved quantities (ρ and $\rho\mathbf{u}$). To obtain orthogonal transformation matrix [23] we use use Gram-Schmidt orthogonalization process and eventually get

$$\mathbb{K}^T = \begin{bmatrix}
1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1\\
0 & -1 & -1 & -1 & 0 & 1 & 1 & 1 & 0\\
0 & 1 & 0 & -1 & -1 & -1 & 0 & 1 & 1\\
-4 & 2 & -1 & 2 & -1 & 2 & -1 & 2 & -1\\
0 & 0 & 1 & 0 & -1 & 0 & 1 & 0 & -1\\
0 & 1 & 0 & -1 & 0 & 1 & 0 & -1 & 0\\
0 & -1 & 0 & 1 & -2 & 1 & 0 & -1 & 2\\
0 & 1 & -2 & 1 & 0 & -1 & 2 & -1 & 0\\
4 & 1 & -2 & 1 & -2 & 1 & -2 & 1 & -2
\end{bmatrix}. \tag{12}$$

The streaming step is defined by

$$f_i(\mathbf{x}+\mathbf{c}_i,\ t+1) = f_i'(\mathbf{x},\ t), \tag{13}$$

and collision step by

$$f_i^{'}(\mathbf{x},\ t) = f_i(\mathbf{x},\ t) + \mathbb{K}\cdot\mathbf{k} + F_i, \tag{14}$$

where $f_i^{'}$ is the so-called post-collision velocity distribution function. The collision vector **k**, has nine components $k_1,\ ...,k_9$, where $k_1,\ k_2,\ k_3$ vanish due to the conservation of density and momenta,

$$\mathbf{k} = [0,0,0,\ k_4,\ k_5,\ k_6,\ k_7,\ k_8,\ k_9]^T. \tag{15}$$

The equilibrium central moments for the fluid flow are defined as

$$\begin{aligned}
\kappa_{00} &= \kappa_{00}^{eq} = \rho,\\
\kappa_{10}^{eq} &= \kappa_{10} = 0,\\
\kappa_{01}^{eq} &= \kappa_{01} = 0,\\
\kappa_{20}^{eq} &= \rho c_s^2,\\
\kappa_{02}^{eq} &= \rho c_s^2,\\
\kappa_{11}^{eq} &= 0,\\
\kappa_{21}^{eq} &= 0,\\
\kappa_{12}^{eq} &= 0,\\
\kappa_{22}^{eq} &= \rho c_s^4.
\end{aligned} \tag{16}$$

Cascaded collision scheme is given by the following equations for components of the **k**

$$\begin{aligned}
k_4 &= \tfrac{1}{12\tau_4}[\rho(u^2+v^2) - f_6 - f_8 - f_4 - f_2 - 2(f_5+f_3+f_7+f_1-\rho c_s^2)\\
&\quad + F_4],\\
k_5 &= \tfrac{1}{4\tau_5}[f_8+f_4-f_6-f_2+\rho(u^2-v^2)+F_5],\\
k_6 &= \tfrac{1}{4\tau_6}(f_7+f_3-f_1-f_5-uv\rho+F_6),\\
k_7 &= -\tfrac{1}{4\tau_7}\Big\{[f_5+f_3-f_7-f_1-2u^2v\rho+v(\rho-f_8-f_4-f_0)]+\\
&\quad + 2u(f_7-f_1-f_5+f_3)+F_7\Big\}+\tfrac{v}{2}(-3k_4-k_5)+2uk_6,\\
k_8 &= -\tfrac{1}{4\tau_8}\Big\{[f_3+f_1-f_5-f_7-2v^2u\rho+u(\rho-f_2-f_6-f_0)]+\\
&\quad + 2v(f_7-f_1-f_5+f_3)+F_8\Big\}+\tfrac{u}{2}(-3k_4-k_5)+2uk_6,\\
k_9 &= \tfrac{1}{4}\Big[\tfrac{1}{\tau_9}(\rho c_s^4-\kappa_{xxyy}) - 8k_4 - 6k_5(u^2+v^2) - 2k_5(v^2-u^2)+\\
&\quad + 16k_6uv - 8k_7v - 8k_8u + F_9\Big]
\end{aligned} \tag{17}$$

where $\tau_{4,5,6,...,9}$ are the relaxation times for non-conserved moments. Relations between relaxation times and a kinematic viscosity of the fluid are following

$$\tau_5 = \tau_6 = \frac{\nu}{c_s^2} + \frac{1}{2}. \tag{18}$$

To achieve zero-slip at walls for half-way bounce back, we also set [46].

$$\tau_7 = \tau_8 = \frac{8\tau_5 - 1}{16\tau_5 - 8}, \tag{19}$$

two last relaxation times are set to $\tau_4 = \tau_9 = 1$. Two principal differences between the cascaded scheme and other MRT schemes [10] are that

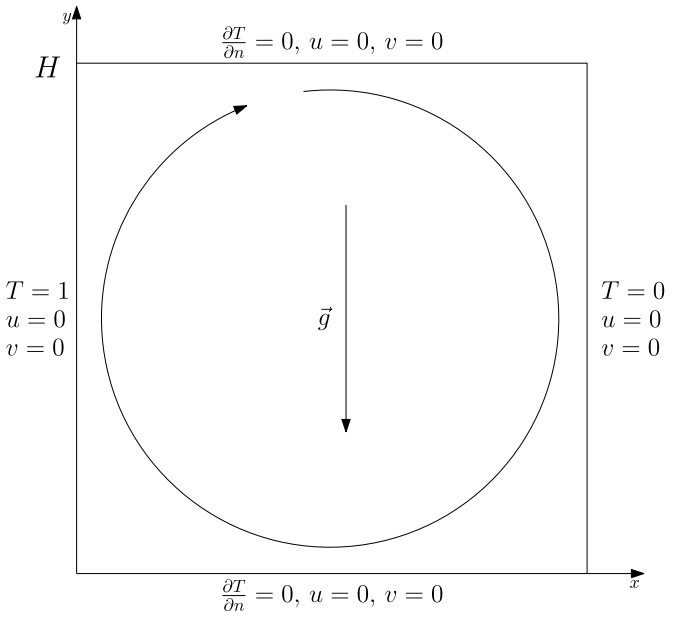


Fig. 2. Differentially heated cavity geometry and boundary conditions.

central moments are relaxed in the cascaded manner [30] and the cross talking among different moments are corrected [23,25]. In order to recover the Navier-Stokes equations from the LBE, Chapmann-Enskog expansion [39], Hilbert expansion [26], Equivalent Partial Differential Equations [12] or recently emerged Maxwell iteration method [41] techniques could be used.

In every case we do Taylor expansion of the LBE, apply diffusive scaling and after lengthy algebra we eventually end up with [18,19,25] (with *Ma* being the Mach number)

$$\begin{aligned}
&\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = O(Ma^2)\\
&\frac{\partial u}{\partial t} + \frac{\partial u^2}{\partial x} + \frac{\partial uv}{\partial y} = -\frac{\partial p}{\partial x} + \frac{1}{Re}\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) + \Gamma_x + O(Ma^3)\\
&\frac{\partial v}{\partial t} + \frac{\partial v^2}{\partial y} + \frac{\partial vu}{\partial x} = -\frac{\partial p}{\partial y} + \frac{1}{Re}\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}\right) + \Gamma_y + O(Ma^3).
\end{aligned} \tag{20}$$

2.2. Cascaded LBM for the temperature field

Temperature is considered as a passive scalar obeying the advection-diffusion equation. Viscous heat dissipation and compressive work have not been considered here as their contribution in natural convection can be neglected in the cases we studied. The CTLBM is used with the D_2Q_5 lattice model. The evolution of temperature is defined by "temperature" distribution functions g_i. The discretized cascaded thermal lattice Boltzmann equation in lattice units reads

$$g(\mathbf{x} + \mathbf{c}_i,\ t + 1) = g(\mathbf{x},\ t) + \mathbb{Q}\cdot\mathbf{q}, \tag{21}$$

where $\mathbb{Q}\cdot\mathbf{q}$ is the cascaded collision scheme for the temperature field, which is similar to the collision scheme used for fluid flow but applied to temperature DFs on the separate lattice. $\mathbb{Q}$ is the orthogonal transformation matrix, and $\mathbf{q}$ is the collision vector. The streaming and collision processes are similar as before with streaming along the links and collision at nodes. The equilibrium DFs for temperature reads

$$\begin{bmatrix} g_1^{eq} \\ g_2^{eq} \\ g_3^{eq} \\ g_4^{eq} \\ g_5^{eq} \end{bmatrix} = \begin{bmatrix} (1-a)T \\ (a-2u)\frac{T}{4} \\ (a-2v)\frac{T}{4} \\ (a+2u)\frac{T}{4} \\ (a+2v)\frac{T}{4} \end{bmatrix}, \tag{22}$$

Table 1
Natural convection in a square cavity: grid convergence for Ma = 0.01.

N	$\lvert T\rvert_{L_\infty}$	$\lvert P\rvert_{L_\infty}$	$\lvert \mathbf{u}\rvert_{L_2}$
Ra= 10^3			
64	$1.745\cdot10^{-4}$	$1.101\cdot10^{-2}$	$1.670\cdot10^{-3}$
128	$4.437\cdot10^{-5}$	$2.730\cdot10^{-3}$	$4.110\cdot10^{-4}$
256	$1.120\cdot10^{-5}$	$9.000\cdot10^{-4}$	$9.663\cdot10^{-5}$
512	$2.578\cdot10^{-6}$	$2.768\cdot10^{-4}$	$1.957\cdot10^{-5}$
768	$9.201\cdot10^{-7}$	$9.085\cdot10^{-5}$	$5.593\cdot10^{-6}$
n	2.0908	1.8583	2.2587
Ra= 10^4			
64	$6.730\cdot10^{-4}$	$3.645\cdot10^{-2}$	$2.529\cdot10^{-3}$
128	$1.608\cdot10^{-4}$	$8.018\cdot10^{-3}$	$6.418\cdot10^{-4}$
256	$3.710\cdot10^{-5}$	$2.478\cdot10^{-3}$	$1.528\cdot10^{-4}$
512	$7.464\cdot10^{-6}$	$7.429\cdot10^{-4}$	$3.113\cdot10^{-5}$
768	$2.272\cdot10^{-6}$	$2.421\cdot10^{-4}$	$8.474\cdot10^{-6}$
n	2.2627	1.9425	2.2536
Ra= 10^5			
64	$2.597\cdot10^{-3}$	$9.753\cdot10^{-2}$	$6.796\cdot10^{-3}$
128	$6.448\cdot10^{-4}$	$1.837\cdot10^{-2}$	$1.803\cdot10^{-3}$
256	$1.501\cdot10^{-4}$	$3.910\cdot10^{-3}$	$4.338\cdot10^{-4}$
512	$2.974\cdot10^{-5}$	$9.645\cdot10^{-4}$	$8.784\cdot10^{-5}$
768	$7.796\cdot10^{-6}$	$2.987\cdot10^{-4}$	$2.289\cdot10^{-5}$
n	2.2954	2.2787	2.2494
Ra= 10^6			
64	$7.983\cdot10^{-3}$	$2.202\cdot10^{-1}$	$2.400\cdot10^{-2}$
128	$2.131\cdot10^{-3}$	$4.806\cdot10^{-2}$	$6.172\cdot10^{-3}$
256	$5.182\cdot10^{-4}$	$8.384\cdot10^{-3}$	$1.498\cdot10^{-3}$
512	$1.034\cdot10^{-4}$	$1.460\cdot10^{-3}$	$3.004\cdot10^{-4}$
768	$2.683\cdot10^{-5}$	$3.945\cdot10^{-4}$	$7.718\cdot10^{-5}$
n	2.2502	2.5284	2.2639
Ra= 10^7			
64	$2.987\cdot10^{-2}$	$4.308\cdot10^{-1}$	$1.200\cdot10^{-1}$
128	$7.133\cdot10^{-3}$	$1.247\cdot10^{-1}$	$3.080\cdot10^{-2}$
256	$1.861\cdot10^{-3}$	$2.255\cdot10^{-2}$	$7.688\cdot10^{-3}$
512	$4.551\cdot10^{-4}$	$3.888\cdot10^{-3}$	$1.827\cdot10^{-3}$
768	$1.864\cdot10^{-4}$	$1.431\cdot10^{-3}$	$7.374\cdot10^{-4}$
1024	$9.227\cdot10^{-5}$	$6.943\cdot10^{-4}$	$3.577\cdot10^{-4}$
n	2.0606	2.3642	2.0824

where T is the temperature and a is free parameter related to the "speed of sound" (not the real one but linked to the lattice). We set $a = \sqrt{2/5}$ [46]. The moments for the temperature field for the D_2Q_5 lattice are

$$\left\{\sum_i g_i,\ c_{i,x}g_i,\ c_{i,y}g_i,\ (c_{i,x}^2 + c_{i,y}^2)g_i,\ (c_{i,y}^2 - c_{i,x}^2)g_i\right\}.$$

The first moment $T = \sum_i g_i$ is the only conserved quantity (collisional invariant). As in previous section, we apply Gram-Schmidt orthogonalization process to obtain the orthogonal transformation matrix $\mathbb{Q}$

$$\mathbb{Q}^{\mathrm{T}} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & -1 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 & 1 \\ 4 & -1 & -1 & -1 & -1 \\ 0 & -1 & 1 & -1 & 1 \end{bmatrix}. \tag{23}$$

The collision step is given by

$$g_i^{'}(\mathbf{x},\ t) = g_i(\mathbf{x},\ t) + \mathbb{Q}\cdot\mathbf{q}, \tag{24}$$

and streaming step for CTLBM reads

$$g_i(\mathbf{x} + \mathbf{c}_i,\ t + 1) = g_i^{'}(\mathbf{x},\ t), \tag{25}$$

where $g_i^{'}$ are post collision DFs.

The cascaded relaxation scheme for the temperature DFs reads

Table 2
Natural convection in a square cavity: grid convergence for Ma = 0.05.

N	$\|T\|_{L_\infty}$	$\|P\|_{L_\infty}$	$\|\mathbf{u}\|_{L_2}$
Ra= 10^3			
64	$2.125\cdot10^{-4}$	$5.645\cdot10^{-3}$	$1.724\cdot10^{-3}$
128	$6.274\cdot10^{-5}$	$2.035\cdot10^{-3}$	$4.221\cdot10^{-4}$
256	$1.932\cdot10^{-5}$	$8.196\cdot10^{-4}$	$1.011\cdot10^{-4}$
512	$5.579\cdot10^{-6}$	$2.725\cdot10^{-4}$	$2.118\cdot10^{-5}$
768	$1.912\cdot10^{-6}$	$9.070\cdot10^{-5}$	$6.096\cdot10^{-6}$
n	1.8515	1.5992	2.2336
Ra= 10^4			
64	$6.592\cdot10^{-4}$	$1.533\cdot10^{-2}$	$2.730\cdot10^{-3}$
128	$1.618\cdot10^{-4}$	$5.414\cdot10^{-3}$	$6.898\cdot10^{-4}$
256	$4.130\cdot10^{-5}$	$2.155\cdot10^{-3}$	$1.746\cdot10^{-4}$
512	$9.769\cdot10^{-6}$	$7.073\cdot10^{-4}$	$3.976\cdot10^{-5}$
768	$2.925\cdot10^{-6}$	$2.354\cdot10^{-4}$	$1.152\cdot10^{-5}$
n	2.134	1.6176	2.1557
Ra= 10^5			
64	$2.660\cdot10^{-3}$	$3.368\cdot10^{-2}$	$7.557\cdot10^{-3}$
128	$6.305\cdot10^{-4}$	$7.985\cdot10^{-3}$	$1.930\cdot10^{-3}$
256	$1.449\cdot10^{-4}$	$2.653\cdot10^{-3}$	$4.813\cdot10^{-4}$
512	$2.893\cdot10^{-5}$	$8.254\cdot10^{-4}$	$1.061\cdot10^{-4}$
768	$7.650\cdot10^{-6}$	$2.715\cdot10^{-4}$	$2.947\cdot10^{-5}$
n	2.3107	1.8631	2.1866
Ra = 10^6			
64	$8.858\cdot10^{-3}$	$8.786\cdot10^{-2}$	$2.468\cdot10^{-2}$
128	$2.208\cdot10^{-3}$	$1.667\cdot10^{-2}$	$6.323\cdot10^{-3}$
256	$5.195\cdot10^{-4}$	$3.695\cdot10^{-3}$	$1.523\cdot10^{-3}$
512	$1.034\cdot10^{-4}$	$9.342\cdot10^{-4}$	$3.202\cdot10^{-4}$
768	$2.686\cdot10^{-5}$	$2.939\cdot10^{-4}$	$8.654\cdot10^{-5}$
n	2.2894	2.2408	2.2322
Ra = 10^8			
64	$1.228\cdot10^{-1}$	$6.947\cdot10^{-1}$	$5.363\cdot10^{-1}$
128	$2.744\cdot10^{-2}$	$1.408\cdot10^{-1}$	$2.152\cdot10^{-1}$
256	$6.816\cdot10^{-3}$	$2.539\cdot10^{-2}$	$7.045\cdot10^{-2}$
512	$1.632\cdot10^{-3}$	$5.084\cdot10^{-3}$	$1.722\cdot10^{-2}$
1024	$3.254\cdot10^{-4}$	$9.575\cdot10^{-4}$	$3.411\cdot10^{-3}$
n	2.1191	2.3797	1.8236

$$
\begin{aligned}
q_1 &= 0,\\
q_2 &= \tfrac{1}{2\tau_{g,2}}(\kappa_{10}^{eq} - \kappa_{10}),\\
q_3 &= \tfrac{1}{2\tau_{g,3}}(\kappa_{01}^{eq} - \kappa_{01}),\\
q_4 &= -\tfrac{1}{4\tau_{g,4}}(\kappa_{20}^{eq} - \kappa_{20} + \kappa_{02}^{eq} - \kappa_{02}) - uq_2 - vq_3,\\
q_5 &= \tfrac{1}{4\tau_{g,5}}(\kappa_{20} - \kappa_{20}^{eq} + \kappa_{02}^{eq} - \kappa_{02}) - uq_2 + vq_3,
\end{aligned}
\tag{26}
$$

where $\tau_{g,2}$, $\tau_{g,3}$, $\tau_{g,4}$, $\tau_{g,5}$ are relaxation times for non-conserved thermal moments and again $q_1 = 0$ as it is collisional invariant and corresponds to the conserved quantity. In the cascaded scheme, lower order moments do affect the evolution of the higher order moments as can be seen from (26). The above scheme can be expressed in DFs and known macroscopic quantities as

$$
\begin{aligned}
q_2 &= \tfrac{1}{2\tau_{g,2}}(Tu - g_3 + g_1),\\
q_3 &= \tfrac{1}{2\tau_{g,3}}(Tv - g_4 + g_2),\\
q_4 &= -\tfrac{1}{4\tau_{g,4}}(aT - g_1 - g_2 - g_3 - g_4) + \left(\tfrac{1}{2\tau_{g,4}} - \tfrac{1}{2\tau_{g,2}}\right)u(Tu - g_3 + g_1) +\\
&\quad + \left(\tfrac{1}{2\tau_{g,4}} - \tfrac{1}{2\tau_{g,3}}\right)v(Tv - g_4 + g_2),\\
q_5 &= \tfrac{1}{4\tau_{g,5}}(g_1 + g_3 - g_2 - g_4) + \left(\tfrac{1}{2\tau_{g,5}} - \tfrac{1}{2\tau_{g,2}}\right)u(Tu - g_3 + g_1) +\\
&\quad + \left(\tfrac{1}{2\tau_{g,3}} - \tfrac{1}{2\tau_{g,5}}\right)v(Tv - g_4 + g_2).
\end{aligned}
\tag{27}
$$

Table 3
Natural convection in a square cavity: grid convergence for Ma = 0.1.

N	$\|T\|_{L_\infty}$	$\|P\|_{L_\infty}$	$\|\mathbf{u}\|_{L_2}$
Ra= 10^3			
64	$2.640\cdot10^{-4}$	$4.990\cdot10^{-3}$	$1.757\cdot10^{-3}$
128	$8.756\cdot10^{-5}$	$1.986\cdot10^{-3}$	$4.363\cdot10^{-4}$
256	$3.225\cdot10^{-5}$	$8.263\cdot10^{-4}$	$1.080\cdot10^{-4}$
512	$1.029\cdot10^{-5}$	$2.750\cdot10^{-4}$	$2.380\cdot10^{-5}$
768	$3.456\cdot10^{-6}$	$9.134\cdot10^{-5}$	$6.943\cdot10^{-6}$
n	1.6862	1.5513	2.1856
Ra= 10^4			
64	$6.759\cdot10^{-4}$	$1.277\cdot10^{-2}$	$2.877\cdot10^{-3}$
128	$1.817\cdot10^{-4}$	$5.118\cdot10^{-3}$	$7.626\cdot10^{-4}$
256	$5.275\cdot10^{-5}$	$2.127\cdot10^{-3}$	$2.133\cdot10^{-4}$
512	$1.585\cdot10^{-5}$	$7.069\cdot10^{-4}$	$5.494\cdot10^{-5}$
768	$5.378\cdot10^{-6}$	$2.355\cdot10^{-4}$	$1.679\cdot10^{-5}$
n	1.8947	1.5493	2.0191
Ra= 10^5			
64	$2.634\cdot10^{-3}$	$2.332\cdot10^{-2}$	$7.966\cdot10^{-3}$
128	$6.203\cdot10^{-4}$	$6.706\cdot10^{-3}$	$2.088\cdot10^{-3}$
256	$1.464\cdot10^{-4}$	$2.505\cdot10^{-3}$	$5.606\cdot10^{-4}$
512	$3.120\cdot10^{-5}$	$8.100\cdot10^{-4}$	$1.364\cdot10^{-4}$
768	$8.611\cdot10^{-6}$	$2.685\cdot10^{-4}$	$4.017\cdot10^{-5}$
n	2.2573	1.7234	2.0797
Ra = 10^6			
64	$9.132\cdot10^{-3}$	$5.989\cdot10^{-2}$	$2.515\cdot10^{-2}$
128	$2.222\cdot10^{-3}$	$1.202\cdot10^{-2}$	$6.493\cdot10^{-3}$
256	$5.219\cdot10^{-4}$	$3.089\cdot10^{-3}$	$1.660\cdot10^{-3}$
512	$1.048\cdot10^{-4}$	$8.712\cdot10^{-4}$	$3.844\cdot10^{-4}$
768	$2.741\cdot10^{-5}$	$2.818\cdot10^{-4}$	$1.104\cdot10^{-4}$
n	2.2921	2.0921	2.1386

Table 4
Natural convection in a square cavity: grid convergence for Ma = 0.15.*- the reference solution for Ra= 10^3 is the one with N = 768.

N	$\|T\|_{L_\infty}$	$\|P\|_{L_\infty}$	$\|\mathbf{u}\|_{L_2}$
Ra= 10^{3*}			
64	$3.118\cdot10^{-4}$	$4.744\cdot10^{-3}$	$1.782\cdot10^{-3}$
128	$1.127\cdot10^{-4}$	$1.908\cdot10^{-3}$	$4.453\cdot10^{-4}$
256	$4.163\cdot10^{-5}$	$7.462\cdot10^{-4}$	$1.086\cdot10^{-4}$
512	$1.017\cdot10^{-5}$	$1.853\cdot10^{-4}$	$1.928\cdot10^{-5}$
n	1.6251	1.5390	2.1626
Ra= 10^4			
64	$7.154\cdot10^{-4}$	$1.198\cdot10^{-2}$	$3.035\cdot10^{-3}$
128	$2.079\cdot10^{-4}$	$5.040\cdot10^{-3}$	$8.515\cdot10^{-4}$
256	$7.111\cdot10^{-5}$	$2.126\cdot10^{-3}$	$2.595\cdot10^{-4}$
512	$2.421\cdot10^{-5}$	$7.073\cdot10^{-4}$	$7.165\cdot10^{-5}$
768	$8.196\cdot10^{-6}$	$2.357\cdot10^{-4}$	$2.247\cdot10^{-5}$
n	1.7324	1.5258	1.9194
Ra= 10^5			
64	$2.613\cdot10^{-3}$	$1.967\cdot10^{-2}$	$8.344\cdot10^{-3}$
128	$6.219\cdot10^{-4}$	$6.295\cdot10^{-3}$	$2.274\cdot10^{-3}$
256	$1.533\cdot10^{-4}$	$2.460\cdot10^{-3}$	$6.525\cdot10^{-4}$
512	$3.485\cdot10^{-5}$	$8.047\cdot10^{-4}$	$1.703\cdot10^{-4}$
768	$1.023\cdot10^{-5}$	$2.678\cdot10^{-4}$	$5.202\cdot10^{-5}$
n	2.1851	1.6602	1.9917
Ra= 10^6			
64	$9.262\cdot10^{-3}$	$4.934\cdot10^{-2}$	$2.566\cdot10^{-2}$
128	$2.232\cdot10^{-3}$	$1.046\cdot10^{-2}$	$6.821\cdot10^{-3}$
256	$5.260\cdot10^{-4}$	$2.888\cdot10^{-3}$	$1.858\cdot10^{-3}$
512	$1.072\cdot10^{-4}$	$8.506\cdot10^{-4}$	$4.635\cdot10^{-4}$
768	$2.845\cdot10^{-5}$	$2.780\cdot10^{-4}$	$1.387\cdot10^{-4}$
n	2.2825	2.0163	2.0520

Table 5
Natural convection in a square cavity: convergence of Nusselt numbers and maximal velocities and their positions for Ma = 0.01.*- values omitted from the fit.

N	u_{max}	y	v_{max}	x	Nu	$Nu_{1/2}$
Ra= 10^3						
64	3.65629	0.81329	3.69244	0.17875	1.10912	1.11825
128	3.65234	0.81330	3.69499	0.17856	1.11344	1.11790
256	3.65076	0.81327	3.69623	0.17846	1.11561	1.11782
512	3.65008	0.81326	3.69684	0.17840	1.11670	1.11780
768	3.64986	0.81325	3.69704	0.17837	1.11706	1.11779
1024	3.64975	0.81325	3.69714	0.17836	1.11724	1.11779
∞	3.64945	0.81324	3.69744	0.17832	1.11778	1.11778
Ra= 10^4						
64	16.17183	0.82244	19.59757	0.11966	2.23155	2.25000
128	16.17603	0.82292	19.61129	0.11930	2.23710	2.24609
256	16.17919	0.82309	19.61943	0.11909	2.24069	2.24513
512	16.18114	0.82316	19.62376	0.11898	2.24269	2.24489
768	16.18185	0.82318	19.62524	0.11894	2.24338	2.24485
1024	16.18222	0.82320	19.62601	0.11892	2.24373	2.24483
∞	16.18340	0.82323	19.62830	0.11886	2.24481	2.24481
Ra= 10^5						
64	34.43644	0.85271	68.82944	0.06626	4.52089*	4.55751
128	34.56981	0.85370	68.67551	0.06604	4.51241	4.53036
256	34.65174	0.85419	68.64584	0.06597	4.51489	4.52379
512	34.69525	0.85441	68.63845	0.06592	4.51774	4.52217
768	34.71022	0.85448	68.63699	0.06590	4.51892	4.52187
1024	34.71780	0.85451	68.63646	0.06589	4.51955	4.52177
∞	34.74050	0.85461	68.63500	0.06585	4.52160	4.52160
Ra= 10^6						
64	66.31031	0.85079	222.45383	0.03807	8.82095	8.82671
128	65.10217	0.84961	220.82891	0.03775	8.97848	9.05081
256	64.87421	0.84966	220.58087	0.03778	8.84599	8.88117
512	64.83188	0.84976	220.55092	0.03778	8.82159	8.83898
768	64.82778	0.84981	220.55138	0.03777	8.81996	8.82861
1024	64.82770	0.84983	220.55338	0.03777	8.82173	8.82605
∞	64.83190	0.84990	220.56600	0.03775	8.82522	8.82522
Ra= 10^7						
64	166.86413*	0.86819	659.33366*	0.02152	17.68625	17.82338
128	154.37172	0.87314	704.65400.	0.02125	16.78495	16.85144
256	149.84663	0.87812	700.20230	0.02128	16.57263	16.60530
512	148.84643	0.87921	699.45001	0.02130	16.52722	16.54344
768	148.68581	0.87934	699.36131	0.02131	16.52130	16.53209
1024	148.63414	0.87937	699.33808	0.02130	16.52006	16.52814
2048	148.59562	0.87937	699.33198	0.02130	16.52031	16.52435
∞	148.58000	0.87940	699.32400	0.02130	16.52310	16.52320

Advection-diffusion equation for the temperature can be recovered by adopting same steps as in previous section and reads

$$\frac{\partial T}{\partial t} + \frac{\partial Tu}{\partial x} + \frac{\partial Tv}{\partial y} = \frac{a}{2}\left(\tau_{g,2} - \frac{1}{2}\right)\frac{\partial^2 T}{\partial x^2} + \frac{a}{2}\left(\tau_{g,3} - \frac{1}{2}\right)\frac{\partial^2 T}{\partial y^2} + O(\Delta t^3). \tag{28}$$

The thermal diffusivities can be identified once the above equation is compared with the Fourier-Kirchhoff equation

$$\alpha_x = \frac{a}{2}\left(\tau_{g,2} - \frac{1}{2}\right), \qquad \alpha_y = \frac{a}{2}\left(\tau_{g,3} - \frac{1}{2}\right). \tag{29}$$

Two other relaxation times are set to

$$\tau_{g,4} = \tau_{g,5} = 12\frac{1 - 2\tau_{g,2}}{1 - 12\tau_{g,2}} \tag{30}$$

in order to eliminate numerical slip [59].

2.3. Governing macroscopic equations

In present paper we study application of CTLBM in simulations of natural convection flows, which can be described by Navier-Stokes equations together with Fourier-Kirchhoff equation. The coupling between thermal and flow fields is due to the buoyancy force term added to the momentum equations. The Boussinesq approximation is employed, i.e., we have linear dependency of density on the temperature

$$\rho = \rho_0[1 + \beta(T - T_{\text{ref}})], \tag{31}$$

where ρ_0 is the reference density (at the reference temperature T_{ref}) and β is the linear isobaric thermal expansion coefficient defined by

$$\beta = -\frac{1}{\rho}\left(\frac{\partial \rho}{\partial T}\right)_p. \tag{32}$$

The problem is then described by the following system of PDEs

$$\begin{aligned} &\nabla\cdot\mathbf{u} = 0, \\ &\tfrac{\partial \mathbf{u}}{\partial t} + \mathbf{u}\cdot\nabla\mathbf{u} = -\tfrac{1}{\rho_0}\nabla p + \nu\Delta\mathbf{u} + \Gamma, \\ &\tfrac{\partial T}{\partial t} + \mathbf{u}\cdot\nabla T = \alpha\Delta T, \end{aligned} \tag{33}$$

where force term Γ is given by

$$\Gamma = \mathbf{g}\beta(T - T_{\text{ref}}), \tag{34}$$

with $\mathbf{g}$ being the vector of gravitational acceleration. In order to compare and characterize natural convection in different systems, we need two dimensionless numbers, the Prandtl number (Pr) and Rayleigh (Ra) or Grashof (Gr) number

$$\text{Pr} = \frac{\nu}{\alpha}, \quad Ra = \frac{\beta|\mathbf{g}|\Delta T L^3}{\nu\alpha}, \quad \text{Gr} = \frac{\text{Ra}}{\text{Pr}}, \tag{35}$$

where L is characteristic length of the system and ΔT is temperature difference (e.g. between hot and cold walls). The characteristic velocity

Table 6
Natural convection in a square cavity: convergence of Nusselt numbers and maximal velocities and their positions for Ma = 0.05.

N	u_{max}	y	v_{max}	x	Nu	$Nu_{1/2}$
Ra= 10^3						
64	3.65665	0.81328	3.69282	0.17875	1.10913	1.11822
128	3.65250	0.81329	3.69526	0.17856	1.11343	1.11789
256	3.65087	0.81327	3.69648	0.17846	1.11559	1.11782
512	3.65016	0.81325	3.69708	0.17840	1.11668	1.11780
768	3.64994	0.81324	3.69729	0.17838	1.11704	1.11780
1024	3.64983	0.81324	3.69739	0.17837	1.11722	1.11780
∞	3.64952	0.81322	3.69769	0.17834	1.11776	1.11780
Ra= 10^4						
64	16.17599	0.82239	19.60140	0.11965	2.23160	2.25005
128	16.17785	0.82289	19.61313	0.11930	2.23709	2.24608
256	16.18028	0.82307	19.62094	0.11909	2.24066	2.24510
512	16.18190	0.82315	19.62516	0.11898	2.24265	2.24485
768	16.18250	0.82318	19.62661	0.11894	2.24334	2.24480
1024	16.18281	0.82319	19.62734	0.11892	2.24369	2.24479
∞	16.18380	0.82323	19.62950	0.11886	2.44760	2.24476
Ra= 10^5						
64	34.41572	0.85241	68.82834	0.06623	4.52098	4.55757
128	34.56839	0.85362	68.67820	0.06604	4.51233	4.53026
256	34.65212	0.85415	68.64821	0.06596	4.51480	4.52369
512	34.69677	0.85439	68.64098	0.06591	4.51764	4.52207
768	34.71213	0.85447	68.63961	0.06590	4.51882	4.52177
1024	34.71990	0.85451	68.63911	0.06589	4.51946	4.52167
∞	34.74360	0.85462	68.63800	0.06586	4.52153	4.52153
Ra= 10^6						
64	66.04260	0.84973	222.11650	0.03804	8.98358	9.05594
128	65.08150	0.84926	220.81599	0.03774	8.84599	8.88116
256	64.87265	0.84952	220.59140	0.03778	8.82147	8.83886
512	64.83385	0.84971	220.56106	0.03778	8.81984	8.82849
768	64.83106	0.84979	220.56111	0.03777	8.82083	8.82658
1024	64.83161	0.84983	220.56288	0.03777	8.82160	8.82592
∞	64.83940	0.84994	220.57600	0.03774	8.82508	8.82509
Ra= 10^8						
256	364.92262	0.93898	2236.44327	0.01201	30.64091	30.70105
512	329.87857	0.93039	2224.82989	0.01199	30.31308	30.34281
1024	323.67959	0.92808	2222.93528	0.01200	30.23916	30.25396
2048	322.47709	0.92791	2222.57050	0.01200	30.22461	30.23200
∞	321.61300	0.92790	2222.48000	0.01200	30.22360	30.22460

Table 7
Natural convection in a square cavity: convergence of Nusselt numbers and maximal velocities and their positions for Ma = 0.1.

N	u_{max}	y	v_{max}	x	Nu	$Nu_{1/2}$
Ra= 10^3						
64	3.65710	0.81326	3.69364	0.17875	1.10908	1.11821
128	3.65281	0.81326	3.69604	0.17857	1.11336	1.11790
256	3.65113	0.81324	3.69724	0.17847	1.11552	1.11784
512	3.65039	0.81323	3.69784	0.17840	1.11660	1.11783
768	3.65016	0.81322	3.69803	0.17838	1.11696	1.11783
1024	3.65004	0.81322	3.69813	0.17837	1.11714	1.11783
∞	3.64971	0.81321	3.69842	0.17833	1.11768	1.11783
Ra= 10^4						
64	16.18093	0.82231	19.60712	0.11964	2.23155	2.24998
128	16.18089	0.82285	19.61779	0.11930	2.23698	2.24596
256	16.18246	0.82305	19.62528	0.11909	2.24052	2.24496
512	16.18365	0.82314	19.62936	0.11897	2.24249	2.24470
768	16.18411	0.82317	19.63077	0.11894	2.24318	2.24465
1024	16.18435	0.82318	19.63148	0.11892	2.24353	2.24463
∞	16.18510	0.82322	19.63360	0.11886	2.44600	2.24460
Ra= 10^5						
64	34.41215	0.85219	68.83653	0.06622	4.52068	4.55722
128	34.57100	0.85354	68.68623	0.06603	4.51203	4.52995
256	34.65772	0.85413	68.65651	0.06596	4.51449	4.52338
512	34.70392	0.85440	68.64961	0.06591	4.51734	4.52177
768	34.71980	0.85449	68.64833	0.06589	4.51852	4.52147
1024	34.72782	0.85453	68.64791	0.06588	4.51915	4.52137
∞	34.75230	0.85466	68.64700	0.06584	4.52123	4.52123
Ra= 10^6						
64	65.95183	0.84877	222.06853	0.03802	8.98417	9.05652
128	65.08016	0.84901	220.84664	0.03773	8.84559	8.88075
256	64.88160	0.84941	220.62340	0.03778	8.82108	8.83846
512	64.84803	0.84971	220.59182	0.03777	8.81942	8.82807
768	64.84712	0.84983	220.59150	0.03777	8.82039	8.82615
1024	64.84854	0.84989	220.59309	0.03776	8.82116	8.82548
∞	64.86070	0.85009	220.60600	0.03773	8.82460	8.82460

Table 8
Natural convection in a square cavity: convergence of Nusselt numbers and maximal velocities and their positions for Ma = 0.15.

N	u_{max}	y	v_{max}	x	Nu	$Nu_{1/2}$
Ra= 10^3						
64	3.65770	0.81322	3.69494	0.17876	1.10898	1.11822
128	3.65327	0.81323	3.69730	0.17858	1.11324	1.11794
256	3.65152	0.81321	3.69849	0.17848	1.11539	1.11788
512	3.65075	0.81319	3.69908	0.17842	1.11647	1.11788
768	3.65051	0.81318	3.69928	0.17839	1.11683	1.11788
∞	3.65005	0.81316	3.69967	0.17834	1.11755	1.11789
Ra= 10^4						
64	16.18671	0.82223	19.61542	0.11963	2.23138	2.24981
128	16.18484	0.82280	19.62520	0.11930	2.23675	2.24574
256	16.18555	0.82302	19.63237	0.11909	2.24027	2.24472
512	16.18630	0.82312	19.63629	0.11897	2.24224	2.24445
768	16.18662	0.82316	19.63765	0.11893	2.24292	2.24440
1024	16.18678	0.82317	19.63833	0.11892	2.24327	2.24438
∞	16.18730	0.82322	19.64040	0.11886	2.24433	2.24434
Ra= 10^5						
64	34.41439	0.85201	68.85101	0.06620	4.52017	4.55667
128	34.57930	0.85348	68.70022	0.06601	4.51153	4.52943
256	34.66913	0.85414	68.67088	0.06595	4.51399	4.52287
512	34.71688	0.85444	68.66425	0.06590	4.51683	4.52126
768	34.73329	0.85453	68.66309	0.06588	4.51801	4.52097
1024	34.74155	0.85458	68.66270	0.06587	4.51865	4.52087
∞	34.76680	0.85471	68.66200	0.06583	4.52100	4.52100
Ra= 10^6						
64	65.91483	0.84796	222.09400	0.03801	8.98376	9.05609
128	65.09141	0.84874	220.90094	0.03773	8.84490	8.88005
256	64.90330	0.84938	220.67544	0.03777	8.82039	8.83776
512	64.87526	0.84979	220.64249	0.03777	8.81870	8.82735
768	64.87624	0.84994	220.64178	0.03776	8.81966	8.82542
1024	64.87871	0.85002	220.64318	0.03776	8.82043	8.82475
∞	64.89440	0.85027	220.65600	0.03773	8.82380	8.82380

Table 9
Natural convection in a square cavity: Mach number influence on Nu and $\mathbf{u}_{\max}$.

Ma	u_{max}	y	v_{max}	x	Nu	$Nu_{1/2}$
Ra= 10^3						
0.01	3.64945	0.81324	3.69744	0.17832	1.11778	1.11778
0.05	3.64952	0.81322	3.69769	0.17834	1.11776	1.11780
0.1	3.64971	0.81321	3.69842	0.17833	1.11768	1.11783
0.15	3.65005	0.81316	3.69967	0.17834	1.11755	1.11789
Ra= 10^4						
0.01	16.1834	0.823230	19.6283	0.118860	2.24481	2.24481
0.05	16.1838	0.823231	19.6295	0.118862	2.44760	2.24476
0.1	16.1851	0.823223	19.6336	0.118862	2.44600	2.24460
0.15	16.1873	0.823223	19.6404	0.118856	2.24433	2.24434
Ra= 10^5						
0.01	34.7405	0.854610	68.635	0.0658500	4.52160	4.52160
0.05	34.7436	0.854621	68.638	0.0658644	4.52153	4.52153
0.1	34.7523	0.854656	68.647	0.0658441	4.52123	4.52123
0.15	34.7668	0.854712	68.662	0.0658313	4.52100	4.52100
Ra= 10^6						
0.01	64.8319	0.849900	220.566	0.0377500	8.82522	8.82522
0.05	64.8394	0.849944	220.576	0.0377445	8.82508	8.82509
0.1	64.8607	0.850091	220.606	0.0377313	8.82460	8.82460
0.15	64.8944	0.850268	220.656	0.0377349	8.82380	8.82380

Table 10
Natural convection in a square cavity: number of iterations needed to reach steady state.

N	Ra= 10^3	Ra= 10^4	Ra= 10^5	Ra= 10^6	Ra= 10^7
Ma= 0.01					
64	663000	1406000	1961000	3570000	7193000
128	1296000	2780000	3810000	6965000	14265000
256	2528000	5410000	7325000	13388000	27631000
512	4910000	10513000	14000000	25557000	53103000
768	7238000	15483000	20408000	37222000	58791000
1024	9522000	13483000	26643000	48564000	76547000
Ma= 0.05					Ra= 10^8
64	141000	302000	602000	1116000	unstable
128	278000	595000	1189000	2198000	4663000
256	542000	1163000	2322000	4293000	9481000
512	1056000	2264000	4520000	8348000	17884000
768	1557000	3339000	6664000	12304000	25987000
1024	2051000	4396000	8773000	16193000	33887000
Ma= 0.1					
64	73000	156000	310000	570000	–
128	143000	307000	612000	1130000	–
256	279000	599000	1196000	2211000	–
512	544000	1167000	2331000	4305000	–
768	803000	1722000	3438000	6350000	–
1024	1057000	2268000	4528000	8361000	–
Ma= 0.15					
64	50000	106000	210000	387000	–
128	97000	208000	415000	766000	–
256	190000	407000	812000	1500000	–
512	369000	792000	1582000	2922000	–
768	544000	1169000	2334000	4310000	–
1024	unstable	1540000	3074000	5677000	–

for natural convection flows can be defined as

$$U = \sqrt{\beta|\mathbf{g}|\Delta TL} = \sqrt{\frac{\mathrm{Ra}}{\mathrm{Pr}}}\frac{\nu}{L}. \tag{36}$$

During the computations, the Nusselt number (Nu) is also determined in order to compare our results with other authors. Two Nusselt numbers are computed

$$\mathrm{Nu} = \frac{1}{\Delta T_{\mathrm{lb}} L_{\mathrm{x,lb}}} \sum_{0}^{L_{\mathrm{x,lb}}} \sum_{0}^{L_{\mathrm{y,lb}}} q_x(x, y)\mathrm{d}x\mathrm{d}y, \tag{37}$$

$$\mathrm{Nu}_{1/2} = \frac{1}{\Delta T_{\mathrm{lb}}} \sum_{0}^{L_{\mathrm{y,lb}}} q_x\left(\frac{L_{\mathrm{x,lb}}}{2}, y\right)\mathrm{d}y, \tag{38}$$

Table 11
Natural convection in a square cavity: comparison with literature.

ref.	u_{max}	y	v_{max}	x	$Nu_{1/2}$	Nu/Nu_0
Ra= 10^3						
CTLBM	3.64945	0.81324	3.69744	0.17832	1.11778	1.11778
Le Quéré [33]	3.6494	0.813	3.6974	0.178	1.1178	1.1178†
Wang [17]	3.6494	0.8132	3.6974	0.1783	1.1178	1.1178
Dixit [16]	3.6529	0.8125	3.682	0.17183	1.118	1.121
Kuznik [14]	3.636	0.809	3.686	0.174	–	–
Mezrhab [4]	3.667	–	3.714	–	–	1.112
Dubois [13]	3.649	0.814	3.697	0.176	–	1.117
Li [43]	3.664	0.81	3.699	0.18	–	1.207
D'Orazio [3]	3.6532	0.8125	3.7006	0.1797	–	1.117
Davis [15]	3.649	0.813	3.697	0.178	1.118	1.118
Ra= 10^4						
CTLBM	16.1834	0.82323	19.6283	0.11886	2.24481	2.24481
Le Quéré [33]	16.183	0.823	19.629	0.119	2.245	2.245†
Wang [17]	16.1834	0.8232	19.6278	0.1189	2.2447	2.2447
Dixit [16]	16.163	0.828	19.569	0.125	2.256	2.286
Kuznik [14]	16.167	0.821	19.597	0.12	–	–
Mezrhab [4]	16.202	–	19.644	–	–	2.241
Dubois [13]	16.188	0.822	19.632	0.119	–	2.243
Li [43]	16.351	0.82	19.589	0.12	–	2.2528
D'Orazio [3]	16.237	0.8203	19.6803	0.1172	–	2.235
Davis [15]	16.178	0.823	19.617	0.119	2.243	2.243
Ra= 10^5						
CTLBM	34.7405	0.85461	68.635	0.06585	4.5216	4.52160
Le Quéré [33]	34.75	0.855	68.64	0.066	4.523	4.522†
Wang [17]	34.743	0.8546	68.6318	0.06588	4.5214	4.5213
Dixit [16]	35.521	0.8554	68.655	0.0664	4.519	4.5463
Kuznik [14]	34.962	0.854	68.578	0.067	–	–
Mezrhab [4]	34.805	–	68.63	–	–	4.519
Dubois [13]	34.748	0.856	68.652	0.065	–	4.517
Li [43]	35.703	0.855	68.536	0.065	–	4.535
D'Orazio [3]	34.8225	0.8529	68.7122	0.0637	–	4.504
Davis [15]	34.73	0.855	68.59	0.066	4.519	4.519
Ra= 10^6						
CTLBM	64.8319	0.8499	220.566	0.03775	8.8252	8.8252
Le Quéré [32]	64.83	0.85	220.6	0.038	8.825	8.825†
Wang [17]	64.8277	0.8498	220.5506	0.03779	8.8256	8.8247
Contrino [54]	64.8334	0.8499	220.5644	0.0378	8.8252	8.8252
Dixit [16]	64.186	0.8496	219.866	0.0371	8.5074	8.805
Kuznik [14]	64.133	0.86	220.537	0.038	–	–
Mezrhab [4]	64.793	–	219.663	–	–	8.817
Dubois [13]	64.842	0.849	220.669	0.037	–	8.806
D'Orazio [3]	64.8679	0.8529	221.1869	0.0392	–	8.767
Davis [15]	64.63	0.85	219.36	0.0379	8.799	8.8

Table 12
Natural convection in a square cavity: comparison with literature cont.

ref.	u_{max}	y	v_{max}	x	$Nu_{1/2}$	Nu/Nu_0
Ra= 10^7						
CTLBM	148.58	0.8794	699.324	0.0213	16.5231	16.5232
Le Quéré [32]	148.58	0.879	699.236	0.021	16.523	16.523†
Contrino [54]	148.5852	0.8793	699.3224	0.0213	16.5231	16.5231
Dixit [16]	164.236	0.851	701.922	0.02	16.79†	–
Kuznik [14]	148.768	0.881	702.029	0.02	16.408	–
Mezrhab [4]	148.4	–	998.3	–	–	16.51
Mayne [8]	145.2666	0.8845	703.2526	0.0215	16.3869†	–
Wan [9]	143.56	0.922	714.48	0.022	–	16.656
Ra= 10^8						
CTLBM	321.613	0.9279	2222.48	0.012	30.2236	30.2246
Le Quéré [32]	321.9	0.928	2222.0	0.012	30.225	30.225†
Contrino [54]	321.9063	0.9279	2222.3279	0.012	30.2251	30.2251
Dixit [16]	389.877	0.937	2241.374	0.0112	30.506†	–
Kuznik [14]	321.457	0.94	2243.36	0.121	28.819†	–
Mezrhab [4]	305.332	–	2169.562	–	–	30.033
Mayne [8]	283.0689	0.9455	2223.4424	0.013	29.6256†	–
Wan [9]	296.71	0.93	2259.08	0.012	–	31.486
Markatos [28]	514.3	0.941	1812	0.0135	–	32.045

Table 13
Natural convection in a square cavity: comparison with literature cont.

ref.	u_{max}	y	v_{max}	x	$Nu_{1/2}$	Nu/Nu_0
Ra= 10^9						
CTLBM	491.56	0.911	7040.9422	0.0064	54.8217	54.7531
Le Quéré [31]	–	–	–	–	54.6	–
Dixit [16]	503.24	0.966	6820.07	0.0064	57.35†	–
Barakos [47]	–	–	–	–	–	60.1
Henkes [48]	–	–	–	–	–	59.5
Ra = 10^{10}						
CTLBM	1252	0.97	22290	0.0034	99.836	100.2
Le Quéré [31]	–	–	–	–	100	–
Dixit [16]	2323	0.940233	21463	0.49072	103.663†	–
Markatos [28]	2323	0.9625	16890	0.0055	–	156.85
Barakos [47]	–	–	–	–	–	134.6
Henkes [48]	–	–	–	–	–	133.4

here $L_{x/y,lb}$ are number of lattice sites in x and y direction, q_x is the local heat flux in the horizontal direction, i.e.

$$q_x = \frac{u_{lb} T_{lb}}{\alpha_{lb}} - \frac{\partial T_{lb}}{\partial x} \tag{39}$$

and the temperature derivative is approximated by the finite differences.

In order to determine all parameters needed to perform LBM simulations, we need one more dimensionless number, namely the Mach number (Ma)

$$\mathrm{Ma} = \frac{U}{c_s}. \tag{40}$$

The above system of equations can be rescaled using the following

$$\begin{aligned} \mathbf{x}^+ &= \frac{\mathbf{x}}{L}, \\ t^+ &= \frac{t\alpha}{L^2}, \\ \mathbf{u}^+ &= \frac{\mathbf{u}L}{\alpha}, \\ p^+ &= \frac{pL^2}{\rho_0 \alpha^2}, \\ T^+ &= \frac{T - T_{\mathrm{ref}}}{\Delta T}, \end{aligned} \tag{41}$$

which results in ("+" superscripts are dropped in the following)

$$\begin{aligned} &\nabla\cdot\mathbf{u} = 0, \\ &\tfrac{\partial \mathbf{u}}{\partial t} + \mathbf{u}\cdot\nabla\mathbf{u} = -\tfrac{1}{\rho_0}\nabla p + \mathrm{Pr}\Delta\mathbf{u} + \mathrm{RaPr}T\mathbf{g}_0, \\ &\tfrac{\partial T}{\partial t} + \mathbf{u}\cdot\nabla T = \Delta T, \end{aligned} \tag{42}$$

where $\mathbf{g}_0$ is unit vector in the direction of the gravitational field. To setup parameters in lattice units, needed during the simulations, we use dimensionless numbers defined above to obtain (we can setup either the desired Ma number i.e. the time step and compute lattice viscosity or setup the lattice viscosity directly providing the resulting Ma number is small enough, say less than 0.3)

$$\nu_{lb} = \frac{\mathrm{Ma}c_s}{L_{lb}}\sqrt{\frac{\mathrm{Pr}}{\mathrm{Ra}}}, \quad \alpha_{lb} = \frac{\nu_{lb}}{\mathrm{Pr}} \tag{43}$$

where L_{lb} is the number of lattices used for characteristic length L. The last parameter which must be expressed in lattice units is the $\beta|\mathbf{g}|$

$$(\beta|\mathbf{g}|)_{lb} = \frac{\mathrm{Ra}\nu_{lb}\alpha_{lb}}{L_{lb}^3 \Delta T_{lb}}. \tag{44}$$

Some of the results (especially for Ra numbers greater than 10^9 [31]) are far away from being steady, for such flows the instant values are meaningless as the flow is chaotic. In the theory of turbulent boundary layer, the dimensionless quantities y^+, U^+ and T^+ are used in order to

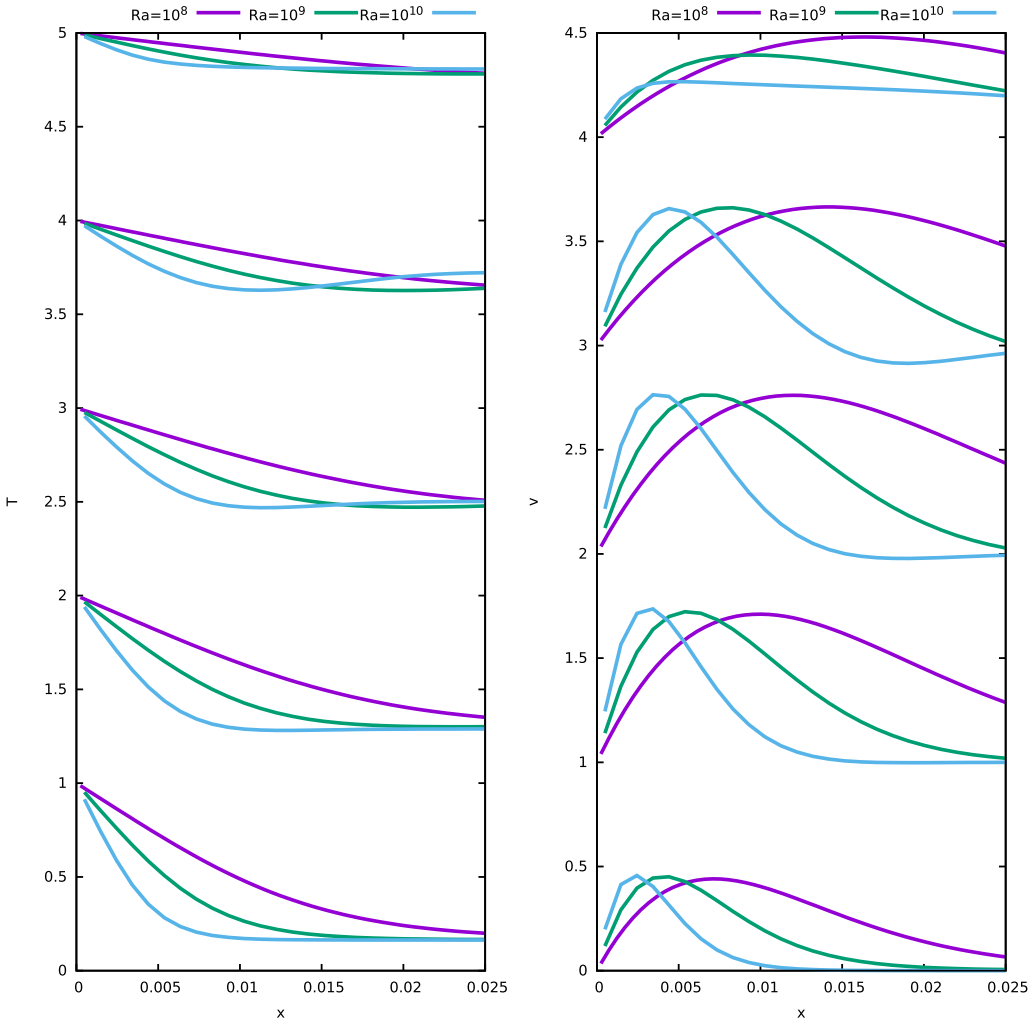


Fig. 3. Averaged temperature (left) and vertical velocity (right) profiles at different positions $y = 0.1,\ 0.3,\ 0.5,\ 0.7,\ 0.9$ and Ra $= 10^8,\ 10^9,\ 10^{10}$.

express the so called law of the wall [38]. These quantities are computed from the normal distance from the wall y and velocity parallel to wall v scaled by the friction velocity v_τ

$$\begin{aligned} y^+ &= \frac{y v_\tau}{\nu}, \\ U^+ &= \frac{v}{v_\tau}, \\ T^+ &= \frac{T_{\text{wall}} - T}{q_w / v_\tau}, \\ v_\tau &= \sqrt{\nu \left(\frac{\partial v}{\partial y}\right)_{y=0}}, \\ q_w &= -\alpha \left(\frac{\partial T}{\partial y}\right)_{y=0}. \end{aligned} \tag{45}$$

The derivatives of v and T are obtained by fitting of the third order polynomial applied to values in the vicinity of the wall.

2.4. Initial and boundary conditions

As the LBM solver is a transitional solver, we need to provide not only the boundary conditions but also initial conditions. The initial conditions are given by zero velocity and reference temperature

$$\begin{aligned} &f_i(\mathbf{x}, 0) = f_i^{\text{eq}}(\rho_{\text{ini,lb}}, \mathbf{u}_{\text{ini,lb}}),\ g_i(\mathbf{x}, 0) = g_i^{\text{eq}}(T_{\text{ini,lb}}, \mathbf{u}_{\text{ini,lb}})\ \forall\ \mathbf{x} \\ &\qquad \in [0, L_{\text{x,lb}}] \times [0, L_{\text{y,lb}}], \\ &\rho_{\text{ini,lb}} = 1,\ \mathbf{u}_{\text{ini,lb}} = (0,0),\ T_{\text{ini,lb}} = 0.5, \end{aligned} \tag{46}$$

where $L_{\text{x,lb}}$ and $L_{\text{y,lb}}$ are lattice dimensions of the computational domain.

Boundary conditions needed during the simulations are no-slip conditions for the velocity at walls, Dirichlet and adiabatic conditions for the temperature at walls. The no-slip and adiabatic conditions can be simulated by the bounce-back approach [40].

$$\begin{aligned} f_{\bar{i}}(\mathbf{x}_{\text{bb}}, t+1) &= f_i'(\mathbf{x}_{\text{bb}}, t), \\ g_{\bar{i}}(\mathbf{x}_{\text{bb}}, t+1) &= g_i'(\mathbf{x}_{\text{bb}}, t), \end{aligned} \tag{47}$$

where $\mathbf{x}_{\text{bb}}$ is the wall adjacent fluid site, $f_{\bar{i}}$ and $g_{\bar{i}}$ correspond to DFs which characteristic velocity points in the reflected direction i.e. $\mathbf{c}_{\bar{i}} = -\mathbf{c}_i$. This results in no-slip and/or adiabatic wall.

In the case of Dirichlet conditions (i.e. for wall with given temperature) we use "anti-bounce-back" conditions to setup the temperature of the walls

$$g_{\bar{i}}(\mathbf{x}_{\text{w}}, t+1) = -g_i(\mathbf{x}_{\text{w}}, t) + \frac{a}{2} T. \tag{48}$$

3. Differentially heated square cavity

The test case for the CTLBM is the well known and well studied

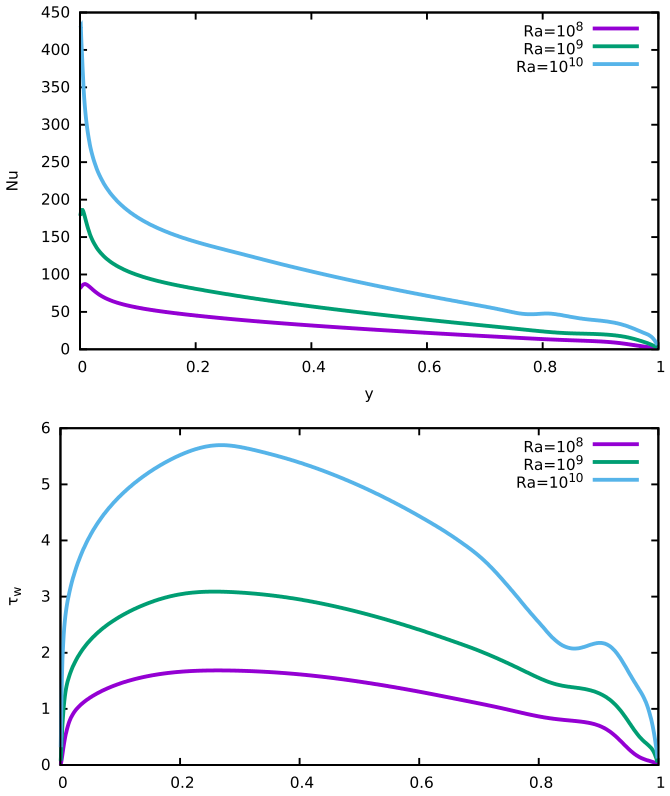


Fig. 4. Local Nusselt number along hot wall (top) and dimensionless wall shear stress (bottom) profiles.

differentially heated square cavity. The cavity's top and bottom walls are insulated, left and right walls are kept at different temperatures (the left wall is the hot wall, see Fig. 2). The fluid inside the cavity is heated up by the hot wall and rise due to buoyancy forces, while it is cooled down by the cold wall and descends which produces various flow patterns inside the cavity. To compare our results with other authors' results we will measure the maximal horizontal and vertical velocity ($u_{\max}$ and $v_{\max}$) at mid-cross-sections (vertical and horizontal ones) together with their positions along that cross-section ($y_{\max}$ and $x_{\max}$) and two Nusselt numbers defined earlier. This problem was extensively studied by many authors [3,9,13–17,28,32,43] and data provided by them will be used for comparison for the Rayleigh numbers up to 10^{10}.

3.1. Numerical simulations setup

The natural convection in a square cavity of height H was simulated for the range of Ra = 10^3 – 10^{10}. The flow in 2D is laminar up to Ra ~ $1.82{\cdot}10^8$ [31], for Ra numbers above Ra = 10^9 the flow is fully chaotic. To setup the LBM solver, we have to choose the Prandtl number (equal to 0.71 in all simulations, i.e. the air filled cavity), the Rayleigh number, and the Mach number. Temperature difference in lattice units was equal to 1 with $T_{\text{hot,lb}} = 1$ and $T_{\text{cold,lb}} = 0$. Initial temperature was set to $T_{\text{ini,lb}} = 0.5$. Initial velocity field is set to zero $\mathbf{u}_{\text{ini,lb}} = (0,0)$. The number of lattice size in x and y direction $L_{x,lb} = L_{y,lb} = N$ were varied from 64 to 1024 (or 2048 for Ra = 10^7 – 10^8). We also performed simulations with different Mach numbers (Ma = 0.01, 0.05, 0.1, 0.15) to examine behavior of quantities under study, when the time step is increased.

For Ra = 10^3, ...,10^8, the computations were performed until the following conditions were met:

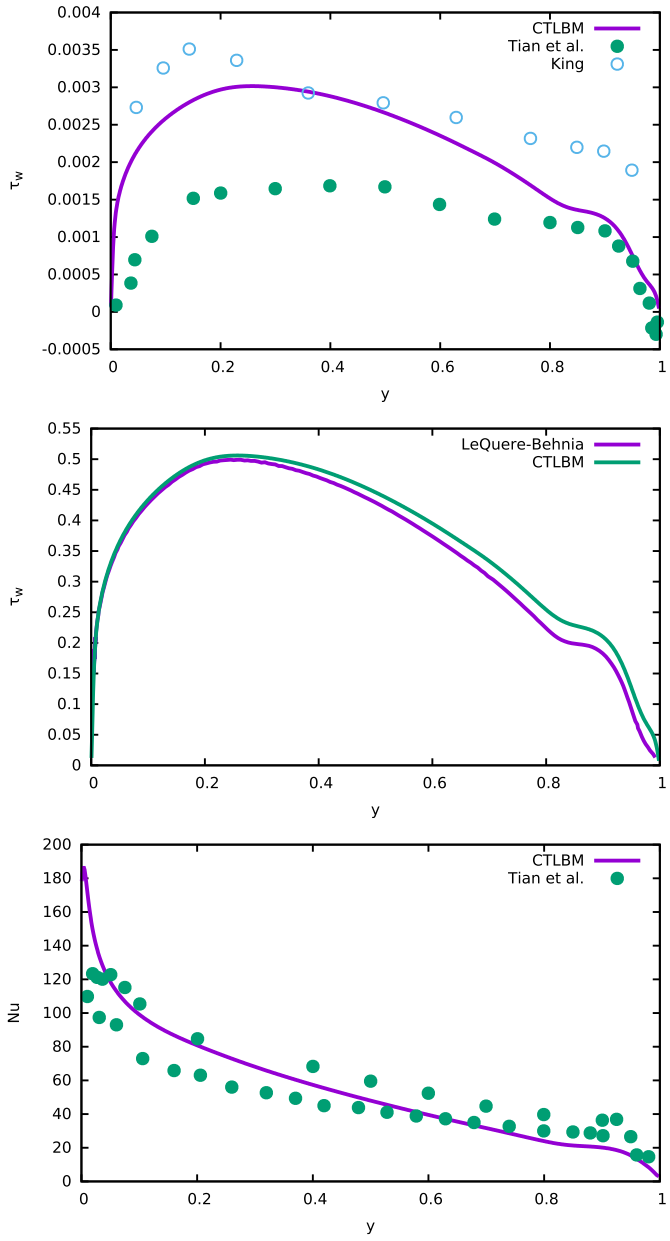


Fig. 5. Wall shear stress and Nusselt number along hot wall for Ra = 10^9 compared to experimental data of Tian et al. [60] and King [61] (top & bottom) and wall shear stress DNS data by Le Quéré and Behnia [31] (center, $\tau_w = \text{Ra}^{-1/4}\partial v/\partial x$).

$$\frac{\sum_{x,y}\left|\mathbf{u}(x,y,t+1000)-\mathbf{u}(x,y,t)\right|_{L_2}}{\sum_{x,y}\left|\mathbf{u}(x,y,t)\right|_{L_2}} < 10^{-10}$$

$$\max_{x,y}|T(x,y,t+1000) - T(x,y,t)| < 10^{-6} \quad (49)$$

i.e. we seek steady state by checking L_2 norm of velocity and L_∞ norm of temperature fields.

3.2. Grid convergence of the solutions

First we check the grid convergence of the solutions for different Ra numbers and Mach numbers. Ma number is related to the time step of the simulations and also to the compressibility error [55]. To measure the errors we compare solutions on several grids with reference solution (grid with $N = 1024$ or in case of Ra = 10^7 and Ra = 10^8 with $N = 2048$). To compute L_2 -velocity and L_∞ -density/temperature norms we

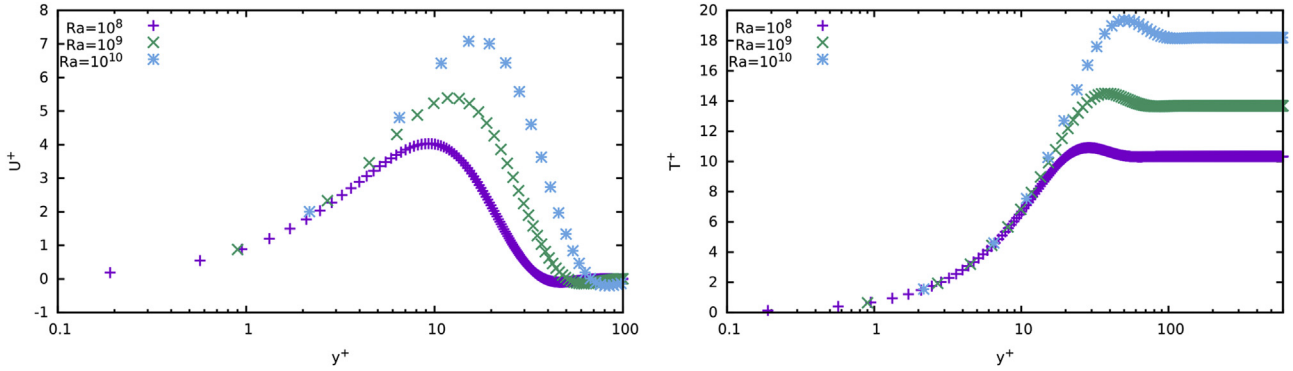


Fig. 6. Dimensionless velocity U^+ (left) and temperature T^+ (right) in wall units y^+ for the three highest Ra numbers.

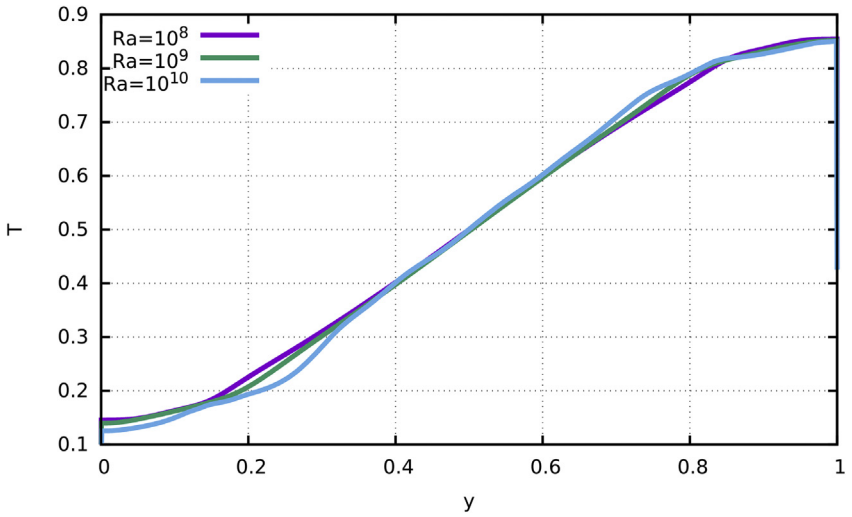


Fig. 7. Averaged temperature stratification along the mid-width of the cavity.

Table 14
Averaged temperature stratification for Ra = 10^8, 10^9, 10^{10}.

Ra	10^8	10^9	10^{10}
S_T	0.9872	0.9996	1.0123

interpolate values from fine grid to the coarse grids using cubic interpolation. Results are summarized in Tables 1–4, where n denotes order of the convergence. From the tables one can observe that for small Mach number the orders are close or above 2.0 except for the pressure field for the two smallest Ra numbers. With increasing Mach number the order of convergence for pressure field becomes lower, with lowest values for low Ra numbers. The errors are also increasing with increased Ra numbers.

3.3. Convergence of hydrodynamical quantities and Nusselt numbers

Next we study behavior of averaged Nusselt numbers $N_{1/2}$ and Nu, maximal velocities in x and y directions u_{max} and v_{max} at vertical and horizontal mid-lines, together with their locations. The values and positions are obtained by first fitting velocity to the quadratic function in the vicinity of node with maximal value and then this function is used to compute maximal values and their positions. Results are summarized in Tables 5–8. Asymptotic values (denoted by "$N = \infty$") are obtained by least-square fit of the 3rd order polynomial function as done in Ref. [17]. Comparison of asymptotic values for different Ma numbers are summarized in Table 9. The most sensitive are components of maximal horizontal and vertical velocity, while their positions are not very sensitive the increasing time step. With increasing Ma, the maximal velocities tend to be over predicted while Nu numbers have opposite trend and are under predicted in the most cases. Asymptotic quantities for the Ma= 0.01 will be compared with results from the literature in the next section.

3.4. Comparison with benchmark solutions and other numerical methods

The asymptotic values computed with Ma= 0.01 are compared with literature values by different authors and methods in Tables 11 and 12. Up to Ra= 10^6 we can observe excellent agreement with precise values by Le Quéré [31–33] obtained by the pseudo spectral method and also with Wang et al. [17] obtained by the incompressible version of MRT LBM. For the Rayleigh numbers above 10^6 our results are still the same or very close to the ones obtained by Le Quéré and other authors. In Table 13 are averaged values for high Ra numbers, obtained in the following way: we performed 10^8 iterations with Ma= 0.05, the results in the table are averaged values from the last $5{\cdot}10^7$ iterations. We obtained different values of velocity components for Ra= 10^9 while their positions and Nusselt numbers are consistent with other authors. For the case of Ra= 10^{10} the situation is the same, except position of the v_{max} reported by Dixit [16] (but this could be typo in the original text). The Nusselt numbers obtained from CTLBM simulation is very close to the values given by Le Quéré and Dixit.

From the computation point of view, the number of iterations needed to achieve steady state prescribed earlier is also of interest. The summary is given in Table 10 and it can be seen from the values there that increasing resolution twice also approximately doubles the iterations count. Situation is the same with increasing Ra number, where

increasing Ra by one order causes double increase in iterations. The numerical code was implemented in NVIDIA CUDA framework [29] and ran on different GPUs. The highest MLUPS (Million Lattice Updates Per Second) was achieved on NVIDIA TESLA K40 and NVIDIA GEFORCE TITAN Z cards; 1056 MLUPS and 869 MLUPS respectively are the average performance of the GPUs.

3.5. Results for high Rayleigh numbers

Averaged vertical velocities and temperature profiles at different vertical positions are compared in Fig. 3. From the figures one can clearly see formation of the thin boundary layers, with steep gradients of velocity and temperature.

The local Nusselt number at hot wall and the wall shear stress distributions for Ra = $10^8 - 10^{10}$ are presented in Fig. 4. The comparison of experimental data by Tian et al. [60] and King [61] are presented in Fig. 5 together with data from Le Quéré and Behnia [31]. We can observe good agreement for both Nusselt number and shear stress values along hot wall.

In Fig. 6 dimensionless velocities and temperatures are reported in wall units defined earlier, we can observe a clue of the universal profile up to $y^+ = 5$ for the vertical velocities and up to $y^+ = 11$ for the temperature within the near-wall region.

Temperature stratification for different Ra numbers is plotted in Fig. 7 and also reported in Table 14, where we give its numerical values defined as

$$S_T = \frac{\partial T}{\partial y}(0.5, 0.5)$$

where T and y are dimensionless.

4. Conclusions

The CTLBM was applied to natural convection in differentially heated cavity. From the presented results it is clear that CTLBM can deliver accurate solutions. The performed grid convergence tests showed that the method is of second order. It turned out that CTLBM gives results which agree with benchmark solutions computed by other authors. For the high Ra regime, the Nusselt number and wall shear stress distributions along the hot wall are in good agreement with experimental and DNS data.

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